are shown as horizontal bars. Figure 6-5 shows a simple Gantt chart for Project X created with Microsoft Project. Figure 6-6 shows a Gantt chart that is more sophisticated and is based on a software launch project from a template provided by Microsoft. Recall that the activities on the Gantt chart should coincide with the activities on the WBS, which should coincide in turn with the activity list and milestone list. Notice that the Gantt chart for the software launch project contains milestones, summary tasks, individual task durations, and arrows showing task dependencies.

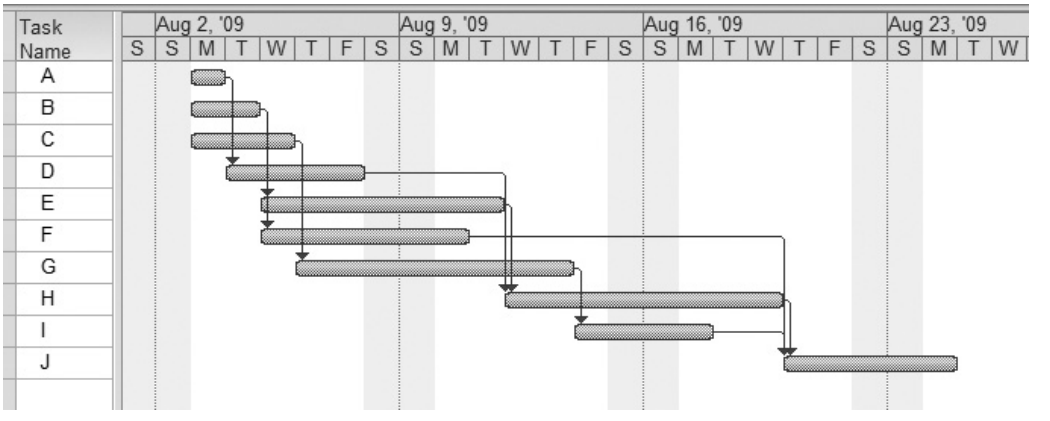

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FIGURE 6-5 Gantt chart for Project X

Notice the different symbols on the Gantt chart for the software launch project (see Figure 6-6):

- The black diamond represents a milestone. In Figure 6-6, Task 1, "Marketing Plan distributed," is a milestone that occurs on March 17. Tasks 3, 4, 8, 9, 14, 25, 27, 43, and 45 are also milestones. For very large projects, top managers might want to see only milestones on a Gantt chart. Microsoft Project allows you to filter information displayed on a Gantt chart so you can easily show specific tasks, such as milestones.
- The thick black bars with arrows at the beginning and end represent summary tasks. For example, Activities 12 through 15—"Develop creative briefs," "Develop concepts," "Creative concepts," and "Ad development"—are all subtasks of the summary task called "Advertising," Task 11. WBS activities are referred to as tasks and subtasks in most project management software.
- The light gray horizontal bars for Tasks 5, 6, 7, 10, 12, 13, 15, 22, 24, 26, and 44 represent the duration of each individual task. For example, the light gray bar for Subtask 5, "Packaging," starts in mid-February and extends until early May.
- Arrows connecting these symbols show relationships or dependencies between tasks. Gantt charts often do not show dependencies, which is their major disadvantage. If dependencies have been established in Microsoft Project, they are automatically displayed on the Gantt chart.

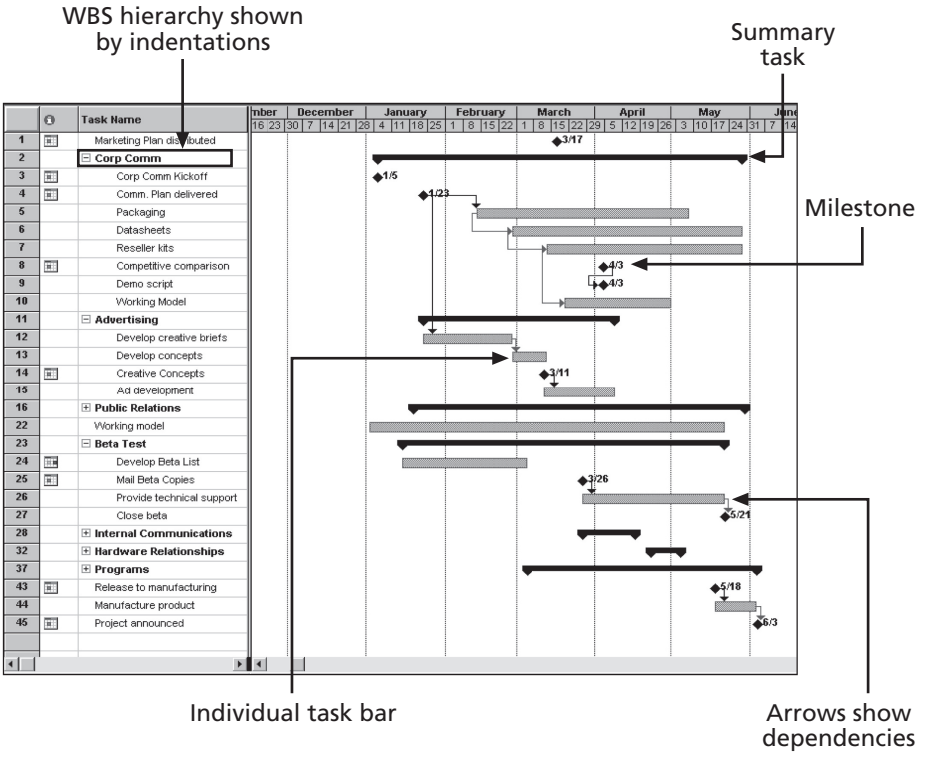

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FIGURE 6-6 Gantt chart for software launch project

Adding Milestones to Gantt Charts

Milestones can be a particularly important part of schedules, especially for large projects. Many people like to focus on meeting milestones, so you can create them to emphasize important events or accomplishments on projects. Normally, you create milestones by entering tasks with zero duration. In Microsoft Project, you can mark any task as a milestone by checking the appropriate box in the Advanced tab of the Task Information dialog box. The duration of the task will not change to zero, but the Gantt chart will show the milestone symbol to represent that task based on its start date. See Appendix A for more information.

To make milestones meaningful, some people use the SMART criteria to help define them. The **SMART criteria** are guidelines suggesting that milestones should be:

- **S**pecific
- **M**easurable
- **A**ssignable
- **R**ealistic
- **T**ime-framed

For example, distributing a marketing plan is specific, measurable, and assignable if everyone knows what should be in the marketing plan, how it should be distributed, how

many copies should be distributed and to whom, and who is responsible for the actual delivery. Distributing the marketing plan is realistic and able to be time-framed if it is an achievable event and scheduled at an appropriate time.

BEST PRACTICE

Schedule risk is inherent in the development of complex systems. Luc Richard, the founder of *www.projectmangler.com*, suggests that project managers can reduce schedule risk through project milestones, a best practice that involves identifying and tracking significant points or achievements in the project. The five key points of using project milestones include the following:

1. Define milestones early in the project and include them in the Gantt chart to provide a visual guide.
2. Keep milestones small and frequent.
3. The set of milestones must be all-encompassing.
4. Each milestone must be binary, meaning it is either complete or incomplete.
5. Carefully monitor the critical path.

Additional best practices that Richard recommends for software development projects include the following:

- Monitor the project's progress and revise the plan.
- Build on a solid base; that means developing a system with less than a 0.1 percent defect rate.
- Assign the right people to the right tasks. Put the best developers on the critical tasks.
- Start with high-risk tasks.
- "Don't boil the ocean." In other words, if the entire project consists of high-risk tasks, then the project itself is a high risk and bound for failure.
- Integrate early and often, and follow practices like the daily build process.[6] (A daily build involves compiling the latest version of a software program each day to ensure that all required dependencies are present and that the program is tested to avoid introducing new bugs.)

Using Tracking Gantt Charts to Compare Planned and Actual Dates

You can use a special form of a Gantt chart to evaluate progress on a project by showing actual schedule information. Figure 6-7 shows a **Tracking Gantt chart**—a Gantt chart that compares planned and actual project schedule information. The planned schedule dates for activities are called the **baseline dates**, and the entire approved planned schedule is called the **schedule baseline**. The Tracking Gantt chart includes columns labeled "Start" and "Finish" to represent actual start and finish dates for each task, as well as columns labeled "Baseline Start" and "Baseline Finish" to represent planned start and finish dates for each task. (These columns are hidden in Figure 6-7.)

In this example, the project is completed, but several tasks missed their planned start and finish dates.

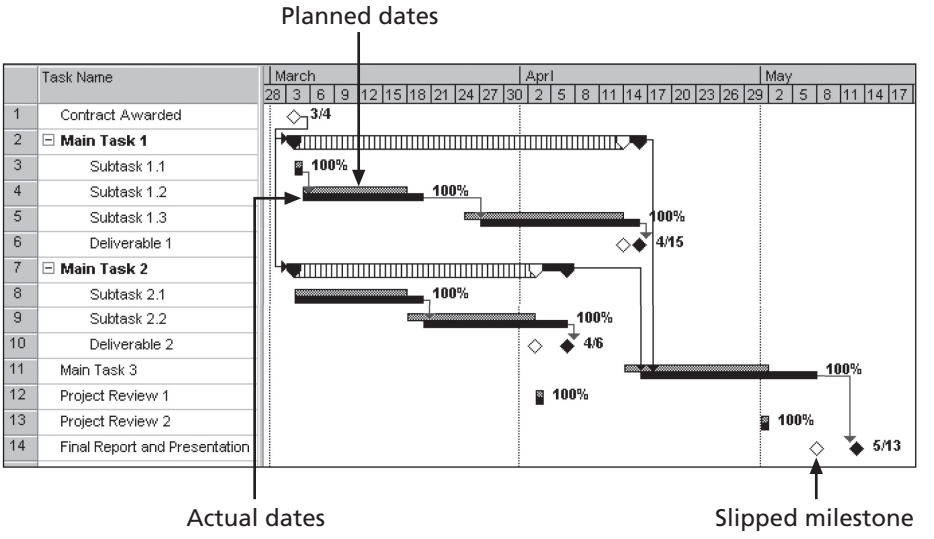

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FIGURE 6-7 Sample Tracking Gantt chart

To serve as a progress evaluation tool, a Tracking Gantt chart uses a few additional symbols:

- Notice that the Gantt chart in Figure 6-7 often shows two horizontal bars for tasks. The top horizontal bar represents the planned or baseline duration for each task. The bar below it represents the actual duration. Subtasks 1.2 and 1.3 illustrate this type of display. If these two bars are the same length, meaning they start and end on the same dates, then the actual schedule was the same as the planned schedule for that task. This scheduling occurred for Subtask 1.1, in which the task started and ended as planned on March 4. If the bars do not start and end on the same dates, then the actual schedule differed from the planned or baseline schedule. If the top horizontal bar is shorter than the bottom one, the task took longer than planned, as you can see for Subtask 1.2. If the top horizontal bar is longer than the bottom one, the task took less time than planned. A striped horizontal bar, as illustrated by Main Tasks 1 and 2, represents the planned duration for summary tasks. The black bar adjoining it shows progress for summary tasks. For example, Main Task 2 clearly shows that the actual duration took longer than planned.
- A white diamond on the Tracking Gantt chart represents a **slipped milestone**. A slipped milestone means the milestone activity was actually completed later than originally planned. The last task provides an example of a slipped milestone because the final report and presentation were completed later than planned.